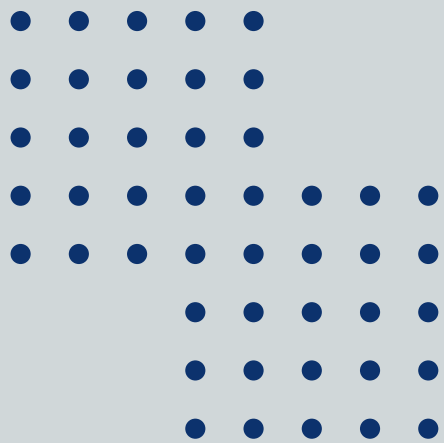
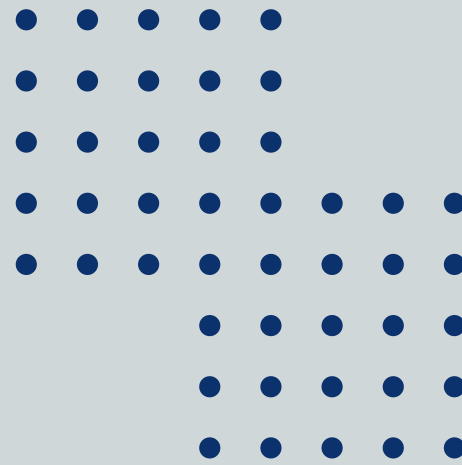




2026

AI IN APARTMENTS.COM

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APARTMENTS.COM

Apartments.com is an online real estate platform owned by Co-Star Group. This platform provides a rental marketplace where people can find apartments, houses, and condos. The platform also provides an option for realtors and property managers to post listings.

OPPORTUNITY

Currently, apartments.com has an AI Smart Search that allows users to search in natural language to set hard-coded filters. However, this does not include being able to search for less clean cut items that are included in apartment descriptions.



ITEMS NOT IN FILTERS

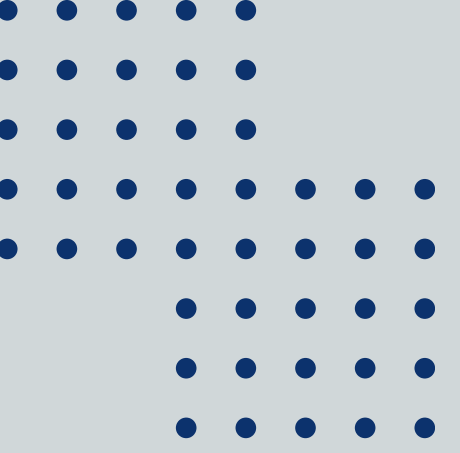
- Hardwood Floors
- Natural Light
- Stainless Steel Appliances
- Spacious Apartment
- Open Layout



PROPOSED SOLUTION



- Enhance smart search for qualitative features
- Extract hard coded filters to use and use apartment description to create vectors for comparison
- Use text analysis to better use the apartment descriptions in listing
- Use cosine similarity to compare the apartments to the user request
- Top apartments passed back through LLM for reranking



DATA ANALYSIS

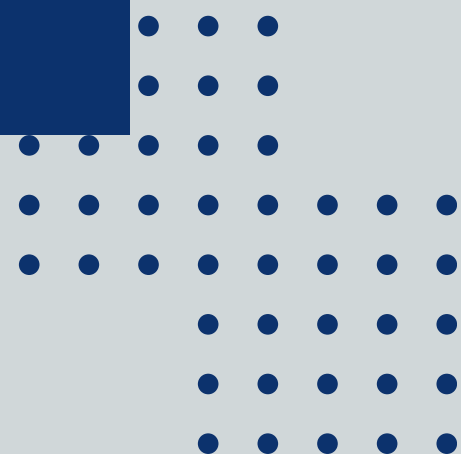
Multi-Stage Recommendation Pipeline

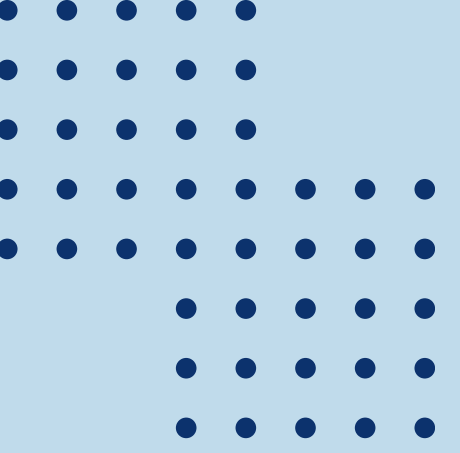
1. LLM Filter Extraction
 - a. Converts natural language into structured filters: rent, bedrooms, amenities, neighborhood
2. Hard Filtering (Baseline)
 - a. Applies traditional filters to reduce dataset
 - b. Ensures results meet core requirements
3. Text-Based Semantic Matching
 - a. Uses cosine similarity
 - b. Compares user input with listing description and reviews
 - c. Captures qualitative preferences

Ranking Method

- Text Similarity Score
- Click Through Rate (CTR)
- Final Score

What We Evaluated

- Number of listings after filtering
 - Ranking differences vs. Traditional Methods
 - Ability to capture “soft” user preferences
- 



EVIDENCE OF SUPPORT

Model Performance Vs. Traditional Filtering

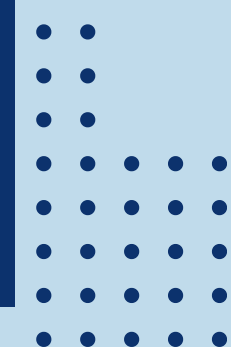
Traditional

- Misses qualitative preferences
- Produces less personalized rankings
- Relevant listings often appear lower in recommendations

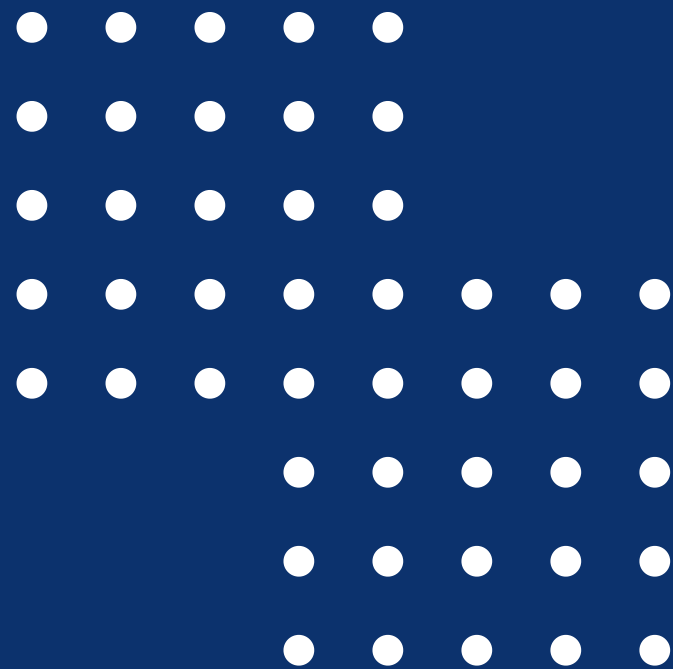
Our Model

- Captures user intent from text (descriptions + reviews)
- Prioritizes stronger matches
- Produces more relevant top results

Findings

- **Soft preferences impact rankings**
 - Listings aligned with user intent positioned higher in rankings where traditional filtering failed to prioritize them
 - **Clear improvements in listing positions in rankings**
 - The most relevant apartments were moved from middle rankings (5-8) under traditional filtering to the top 1-2 results
 - **Improved ordering of listing quality**
 - Apartments with better reviews, descriptions, and engagement (CTR) were ranked higher with our model than traditional filtering
 - **Consistent performance across test prompts**
 - Improvements in ranking observed across three different test prompts with varying user requirements
- 

VALUE CREATED



RENTERS

For tenants, this would make the user experience better as they would be able to search more accurately for apartments. Renters could spend less time searching for apartments on the platform as they would be able to find an apartment faster.



LANDLORDS

Landlords would be able to have their features listed in the description more accurately matched to people looking for the apartment. This would help landlords lease properties faster online without having to pay a separate broker.



APARTMENTS. COM

For [Apartments.com](#), increasing the experience for landlords and users would increase traffic on the website and help the company overall. Using AI would allow Apartments.com to expand their website reach.

LIMITATIONS/EXTENSIONS

Representation Bias

- Data inputted to the model all comes from listings from a realtor
- Inputs come from the people trying to rent the apartments
- Extend inputs to include deep search of the history of the apartment, any issues with the apartment, or any litigation against realtor or landlord

Image Interpretation

- Expand the model to be able to interpret data from images
- Model could rely on images to determine natural lighting, hard wood floors, and other soft categories
- Allows for searches beyond what is included in the description

Geographic Data

- Extend the data to search for how close the apartment is to things like public transportation, gym, grocery store etc.
- This would extend the model to allow users to have more specific requirements, and for landlords to be able to advertise prime locations easier



THANK YOU